

BICYCLE OPTIMIZATION PROJECT – STAGE 3 – ME303

Jennifer Sennett



Manufacturer: Huffy

Model: Rock Creek 26"

Table of Contents

Overview	3
Fatigue Failure - Frame	3
Fatigue Failure - Fork	6
Innovation - Design	8
Reflection	11
References	12
Appendix A.....	12

Overview

The goal of this stage of the project was to determine the maximum rider weight and drop height a bicycle frame and fork could withstand without fatigue failure under infinite life conditions, and to redesign the frame to reduce weight without compromising durability. Fatigue analysis was performed using the Goodman criteria, chosen for its conservative modeling of ductile materials under fluctuating loads. Axial stresses from a 200 lbf rider were calculated for each frame member, identifying the chain stays as the most critical. After applying fatigue modification factors, the endurance limit for infinite life was found to be 21.43 kpsi, and the critical rider weight for fatigue failure was 1250 lbf.

A separate fork analysis modeled repeated drop impacts, using scaled bending and torsional stresses. The maximum drop height before fatigue failure over a million cycles was determined to be 0.405 inches, accounting for stress concentrations and rider load.

To reduce weight, the frame was redesigned with stylized cutouts and reduced wall thickness in low-stress regions. This achieved an 11.55% weight reduction (3.03 lbs to 2.68 lbs), while maintaining safe margins in static and fatigue performance. The results validate that the modified design maintains structural reliability and visually distinctive.

Fatigue Failure - Frame

Methods

The maximum rider weight that the bicycle frame can repeatedly support without fatigue failure, under infinite life conditions, was estimated using the Goodman fatigue criterion. This was implemented through the spreadsheet calculator developed in Stage 1 and 2 of this project.

First, axial stresses in each tube were calculated using known internal forces corresponding to a 200 lbf rider. The member experiencing the highest tensile stress was identified as the critical member, which was most likely to fail first due to fatigue. Modification factors were then applied to estimate the member's endurance limit (S_e), corresponding to infinite life at a million loading cycles. The alternating stress (σ_a) and mean stress (σ_m) for each member were calculated from the maximum and minimum axial stresses (σ_{min} & σ_{max}), and the corresponding Goodman fatigue factor of safety (FOS) was computed in Excel using the calculator from Stage 2.

The Goodman fatigue equation was selected because it is a more conservative approach for ductile metals under fluctuating tensile loads and is analytically simpler for calculations. The rider weight was increased incrementally until the FOS among one of the members under tension

The following equations were used for calculations:

$$\text{Endurance Limit:} \quad S_e = k_a \times k_b \times k_c \times k_d \times k_e \times k_f \times (0.5 \times S_{ut}) \quad (1)$$

$$\text{Surface Condition (K}_a\text{):} \quad a(S_{ut})^b \quad (2)$$

$$\text{Size Modification (K}_b\text{):} \quad 0.879d^{-0.107} \text{ (for diameters between 0.11 in \& 2 in)} \quad (3)$$

$$\text{Alternating Stress } (\sigma_a) \quad \left| \frac{(\sigma_{max} - \sigma_{min})}{2} \right| \quad (4)$$

$$\text{Mean Stress } (\sigma_m) \quad \frac{\sigma_{max} + \sigma_{min}}{2} \quad (5)$$

Goodman Criterion

$$FOS = \frac{1}{\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}}} \quad (6)$$

*Where a and b correspond to the surface finish properties found in Table 6-12 in *Shigley's Mechanical Engineering Design* textbook, and d is the diameter of the member.

Assumptions:

- **All members experience axial stress only; bending and torsion are negligible.** The frame geometry and loading conditions are simplified under the assumption that vertical rider weight produces purely axial forces in the frame members. Additionally, the frame was analyzed using a statics-based model (Method of Joints) where bending moments and torsional effects are not introduced.
- **Tensile axial stress across all members governs fatigue behavior.** Compressive members are ignored as compressive stresses typically do not initiate fatigue.
- **Minimum stress (σ_{min}) is zero.** This is due to loading only occurring when the rider is seated; in the absence of rider weight, the stress is assumed to be zero.
- **Loading is not fully reversible.** The stress cycles from zero to either a maximum tensile, or compressive value (not both). Thus, it cannot be considered reversible as stress does not alternate from maximum tension to an equal maximum compression.
- **The bike frame tubes are not perfect specimens.** Thus, modification factors are required to find an accurate endurance limit to predict infinite life. Modification factors K_d (temperature modification factor), K_e (reliability factor) are assumed to be 1.0 due to ambient operating temperature, standard design reliability, and lack of additional stress concentration effects (all members have no holes/notches, and a uniform cross-sectional area).
- **Stress scales linearly with rider weight.** This is because frame members experience axial loads under vertical rider weight only, which produce stress responses ($\sigma = \frac{Force}{Area}$) that are linear under elastic conditions.

Results

The following results summarize the fatigue-based analysis of the bicycle frame. Using the Stage 1 & 2 Excel calculator, the chain stays were identified as the members with the highest tensile stress relative to cross-sectional area. Using equations (2) and (3), and the cold-drawn steel properties from Table 6-2 in *Shigley's Mechanical Engineering Design* (p. 324), the modification factors for the chain stays were determined. Applying these values in Equation (1) and using an ultimate tensile strength of 60.9 kpsi (from Stage 1), the endurance limit was calculated to be 21.43 kpsi. Under the assumption of non-reversible loading, the minimum stress was zero, and the maximum stress in the critical member was 31.73 kpsi.

The Goodman fatigue criteria (equation 6) was then used to evaluate the factor of safety for all frame members under tensile loading. The rider weight was incremented until the first member reached a fatigue FOS below 1, which is shown in Figure 1 below. This analysis confirmed that the chain stays would fail first, and the critical rider weight for fatigue failure was determined to be 1250 lbf. A summary of results is shown in Table 1 on page 5.

Figure 1.

Excel Calculation for Critical Rider Weight for Fatigue Failure

Input Values		External Reaction Forces (lbf)	
Bicyclist Weight (W) (lbf)	1250	Reaction at A (FA)	597.036
Angles (°)		Reaction at C (FC)	326.482
<BAD	34	*Note 2 FC reactions (2 identical rear supports)	
<ABD	56	Tension = Positive	
<DBC	68	Compression = Negative	
<BCD	55		
<CDB	57		
AB from horizontal	34	Static Analysis	
BC from horizontal	6	Internal Reaction Forces (lbf)	
AD from horizontal	16	FAB (Down Tube)	1855.919
BD from horizontal	74	FAD (Top Tube)	-1600.632
Headtube	16	FBD (Seat Tube)	-3434.360
Member Lengths (in)		FDC (Seat Stays)	473.319
Length AD	20	FBC (Chain Stays)	951.852
Length AB	24	Static Failure Analysis	
Length BC	17	Yield Strength (σ _y , psi)	50800
Length BD	15	Max Allowable Force (σ _{max}) (lbf)	
Length CD	18	Down Tube	6096
Lever Arms		Top Tube	5588
ax	19.897	Seat Tube	3556
bx	4.135	Seat Stays	2540
cx	16.907	Chain Stays	1524
dx	19.225	*Note: Based on cross-sectional area	
Length A to C (ax + cx)	36.804	*Note: FOS = 1 = Yield	
Cross-Sectional Area		Fatigue Failure Analysis	
Top Tube	0.11	Ultimate Tensile Str (S _{ut} , psi)	60900
Seat Tube	0.07	Endurance Limit (S _e , psi)	21433
Seat Stays	0.05	σ _{min}	0
Head Tube	0.07	Frame Members	σ _{max} = σ'
Down Tube	0.12	Down Tube	15466
Chain Stays	0.03	Top Tube	-14551
Critical Bicyclist Weight (lbf)		Seat Tube	-49062
Static:	Does Not Yield	Seat Stays	9466
Fatigue:	1250	Chain Stays	31728
		Critical Member (Tensile):	Chain Stays
		*Note: FOS = 1 = Fatigue Failure	

Table 1.

Summary of Frame Fatigue Failure Results

Modification Factors	Unitless
K _a	0.91
K _b	0.91
K _c	0.85
k _d = k _e = k _f	1
Fatigue Properties	Kpsi
Endurance limit	21.43
Critical σ _{max} (Chain stays)	31.73
Critical Rider Weight (lbf)	1250

Discussion

The fatigue analysis results align with a conservative estimate of the frame's long-term durability, which is characteristic of using the Goodman criterion for fatigue evaluation, as it assumes failure occurs earlier by linearly combining alternating and mean stresses without accounting for material ductility or yielding relief. An alternative that could have been used for a less conservative approach was the Gerber equation due to its more accurate modeling of ductile material behavior under mean stress, however it does not account for yielding and is more complex to implement analytically.

The most critical members of the frame were identified as the chain stays, which is reasonable given they have a relatively small cross-sectional area (smaller than the other members) and their role is in carrying rear axle loads. The calculated endurance limit of 21.43 ksi and appropriate modification factors aligned with typical

fatigue limits for cold-drawn steels, making this result both reasonable and representative of expected material behavior of the frame.

The maximum tensile stress in the chain stays, 31.73 kpsi, is below the material's yield strength but above its endurance limit, placing it within a fatigue-sensitive range. This validates the use of the Goodman equation for predicting infinite-life failure. The estimated critical rider weight of 1250 lbf seems plausible. It suggests the frame can safely support over six times the typical design load of a 200 lbf rider without fatigue failure, assuming ideal conditions, which is reasonable when engineering for infinite life and incorporating conservative safety margins for something as commonly used as a bicycle. This result reflects a robust and conservative frame, though future improvements could consider weld-induced stress concentrations, dynamic effects, or additional off-axis loads for enhanced accuracy.

Fatigue Failure - Fork

Methods

To evaluate the maximum drop height that the bicycle fork could repeatedly endure without fatigue failure, a fatigue analysis was performed using the Goodman criterion. The goal was to determine the maximum impact load that could be applied cyclically without exceeding the fork's endurance limit under infinite life conditions (at least one million cycles). The drop impact over a 0.05-second interval was modeled using the impact force equation found in Stage 2 ($997\sqrt{h}$), which was derived from the momentum force equation and accounts for both impact time and drop height (h), in inches.

First, the endurance limit was estimated and modified using standard fatigue correction factors for infinite life at one million cycles. Stress concentration factors were then determined based on fork geometry by referencing the appropriate charts from *Shigley's Mechanical Engineering Design* for bending and torsion. Bending and torsional stresses were calculated by scaling values from Stage 2, which were originally derived from the reaction force applied to the fork by a 200 lbf rider which was 47.76 lbf.

Using the bending and torsional stresses, the alternating and mean stresses were computed based on the assumption that the loading fluctuates between a minimum value (static rider load) and a maximum value (drop impact). These stresses, combined with the calculated concentration factors, were then used to determine the effective (Von Mises) alternating and mean stresses as functions of drop height.

Finally, these effective stresses were input into the Goodman fatigue equation and solved numerically in MATLAB to determine the maximum drop height before fatigue failure under infinite life. Supporting calculations are provided in Appendix A-2, and the MATLAB code is included in Appendix B-2.

The following equations were used (outside of already provided equations in the Frame Failure section):

Effective Alternating Stress (σ'_a)

$$\left[\left[(Kf)_{bending}(\sigma a)_{bending} + (Kf)_{axial} \frac{(\sigma a)_{axial}}{0.85} \right]^2 + 3[(Kfs)_{torsion}(\tau a)_{torsion}]^2 \right]^{1/2} \quad (7)$$

Effective Mean Stress (σ'_m)

$$\left[\left[(Kf)_{bending}(\sigma m)_{bending} + (Kf)_{axial} \frac{(\sigma m)_{axial}}{0.85} \right]^2 + 3[(Kfs)_{torsion}(\tau m)_{torsion}]^2 \right]^{1/2} \quad (8)$$

Assumptions:

- **Loading is not fully reversible.** The stress cycles from a minimum to a maximum value that is not equal in magnitude. Thus, loading cannot be considered fully reversible stress.

- **Due to combined loading, load medication factor K_c is assumed to be one.** Loading between axial, bending and torsion is accounted for in the effective stress calculations.
- **Stress concentration factors based on circular welded geometry:** The fork is modeled as a circular tube-to-tube welded connection, which justifies using the round shaft with fillet geometry charts from *Shigley's Mechanical Engineering Design* to determine stress concentrations. Welded geometry can assume high notch sensitivity thus k_f equals k_t .

Results

The following results summarize the fatigue-based analysis of the bicycle fork under repeated drop loading. Using the Stage 2 calculated bending and torsional stresses (935 psi and 980 psi respectively) and a baseline fork reaction force R_e of 47.76 lbf (as shown in the free body diagram in Figure 2), linear scaling was applied to estimate impact-induced stresses as a function of drop height.

Stress concentration factors were determined to be 1.7 for k_f bending and 1.45 for k_{fs} torsion from geometric comparisons to round shaft weld fillet charts in *Shigley's Mechanical Engineering Design* Tables A-15-8 and A-15-9. The ultimate tensile strength of the fork material was taken as 60.9 kpsi, and the endurance limit was calculated as 25.22 kpsi using standard modification factors.

The minimum and maximum stress for both bending and torsion were found and expressed as functions of drop height ($19520\sqrt{h}$ & $20452\sqrt{h}$ respectively) and used to calculate the alternating and mean stresses based on equations (4) and (5). Both minimum and maximum values were equal for both bending and torsion and these values are summarized in Table 2 for clarity. These values were then substituted into equations (7) and (8) to determine the effective alternating and mean stresses for use in the Goodman fatigue analysis.

The MATLAB script shown in Appendix B-2 was used to solve the Goodman equation numerically for the drop height and found that the maximum allowable drop height the fork could repeatedly endure without initiating fatigue failure was approximately 0.405 inches. A summary of results is presented in Table 2 on page 8.

Figure 2.

Freebody Diagram of Fork Steerer Interface

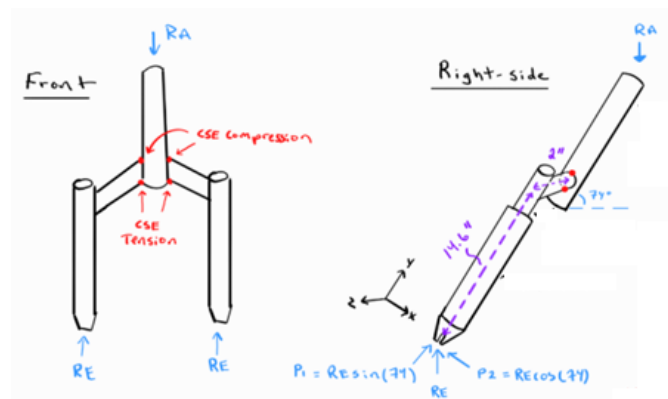


Table 2.*Summary of Max Drop Distance (Infinite Life) Results*

Modification Factors	Unitless
K_a	0.91
K_b	0.91
K_c	1
$k_d = k_e$	1
k_f Bending	1.7
k_f Torsion	1.45
Fatigue Properties	Kpsi
Endurance limit	21.43
$\sigma_{min} = \sigma_{max}$ (Bending)	$19520\sqrt{h}$
$\sigma_{min} = \sigma_{max}$ (Torsion)	$20452\sqrt{h}$
Critical Drop Distance	0.405 in

Discussion

The calculated maximum allowable drop height of 0.405 inches may initially appear small when compared to the fork's static failure capacity of approximately 6.35 ft. However, this result is reasonable within the context of fatigue design for infinite life, meaning it can withstand ≥ 1 million cycles. In fact, when viewed in the context of high-cycle fatigue, a 0.405-inch drop represents a substantial and realistic limit, given that the fork is expected to endure this load over one million cycles without initiating fatigue failure.

The Goodman fatigue equation was applied using relatively high stress concentration factors for both bending and torsion (1.7 and 1.4 respectively) and a nonzero mean stress representing the constant presence of a seated 200 lbf rider. These assumptions significantly reduce the allowable stress amplitude, as both high mean stress levels and stress concentrations can accelerate fatigue damage.

A key understanding for this study is that the goal was not to identify the highest survivable drop height but rather to define the largest drop that can be endured repeatedly without initiating fatigue failure over a long service life. Given these requirements, the small drop height is consistent with conservative design practice and highlights the need for caution when loading structures cyclically. This result also provides a quantitative basis for design improvements if extended drop performance is desired, such as reducing stress concentrations, increasing cross-sectional area, or aiming for an acceptable finite fatigue life instead of infinite life.

Innovation - Design**Description**

To reduce the overall weight of the frame without compromising structural integrity, a series of stylized cutouts were introduced along both the top and down tubes, and the cross-sectional areas of these members were reduced by sixteen and twenty percent, respectively. Finite Element Analysis (FEA) was conducted to ensure these modifications did not significantly affect the frame's strength or safety. A mesh convergence study was performed to validate the accuracy of the FEA results that used a mesh size of 1.15 inches. The mesh convergence is detailed in Appendix B-2 showing that the 1.15 inch mesh size used was adequate. Visual comparisons of the original and redesigned frame geometries are provided in Figures (3 & 4) on page 8 and the FEA results in Figures (5 & 6) the Verification section on page 10.

Figure 3.

CAD image of Original Frame

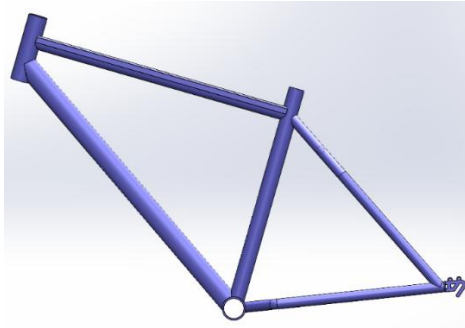
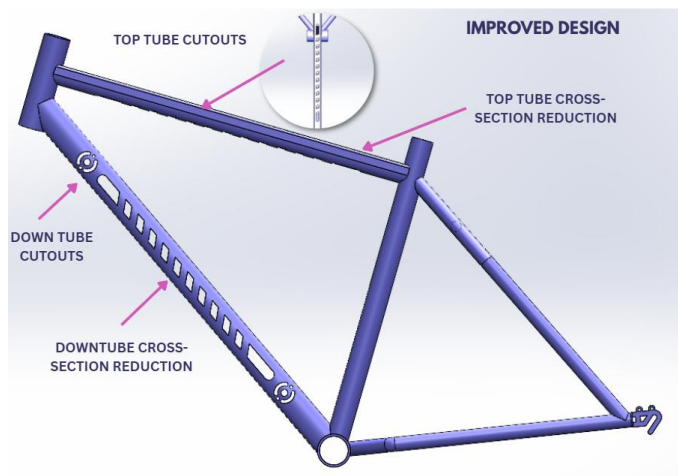
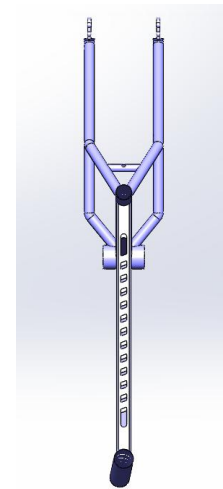


Figure 4.

CAD Images of Redesigned Frame



Side Profile



Top Profile

Verification

The redesigned frame achieves an 11.55% weight reduction, decreasing from 3.03 lbs to 2.68 lbs, as determined using SolidWorks mass properties. To verify that this reduction did not compromise structural integrity, Stage 2 (static) and Stage 3 (fatigue) analyses were rerun using the project's Excel calculator. The factor of safety for yield decreased modestly in both the top and down tubes—from 21.82 to 17.36 and 20.53 to 17.12, respectively—while remaining well above acceptable limits. The critical rider weight for yielding decreased by only 5 lbf—from 1300 lbf to 1295 lbf.

For fatigue analysis, only the down tube was evaluated since the top tube experiences compression and is not a likely subject to fatigue-related failure. The fatigue FOS for the down tube decreased from 24.61 to 20.51, and the overall fatigue critical rider weight for infinite life remained unchanged at 1250 lbf as this design change did not affect the critical member (the chain stays), indicating no loss in long-term durability.

Finite element analysis (FEA) further validates the strength of the redesigned frame. A comparison of von Mises stress distributions between the original frame and the redesigned frame (Figures 5 and 6) shows that the maximum stress increased from 10.9 ksi to 14.0 ksi, which is a slight increase yet still well below the yield

strength of 50.7 ksi. In the top tube, stress rose modestly—up to approximately 4 ksi near the cutouts. The downtube saw the greatest increase with a stress of 14 ksi near circular cutout closest to the head tube and up approximately 7.0 ksi near the junction to the seat tube, however, these increases remain well within safe limits and do not indicate the presence of critical stress concentrations that would cause alarm for yield or fatigue failure. While the gradient shift is visually noticeable, the changes are structurally insignificant, confirming that the material reductions have a minimal impact on overall durability.

Assumptions:

- Loading remains consistent between original and redesigned frames. Primary loading of an average 200 lbf rider is unchanged in both cases allowing for direct comparison.
- Material properties remain unchanged between original and redesigned frame. No material substitutions were made, and both frames have the same surface finish and manufacturing processes to allow direct comparison. Additionally, this aligns with the project requirement to not change frame material.
- Members in compression excluded for fatigue analysis. Objects in compression typically do not fail due to fatigue before those in tension, thus members in tension are more critical to examine.
- Simple boundary conditions are used in FEA simulation. The goal is to compare performance changes rather than predict exact real-world responses, thus simplifying boundary conditions are sufficient and easier to capture critical stress trends and regions.

Figure 5.

Finite Element Analysis Result for Original Frame (Von Mises Stresses)

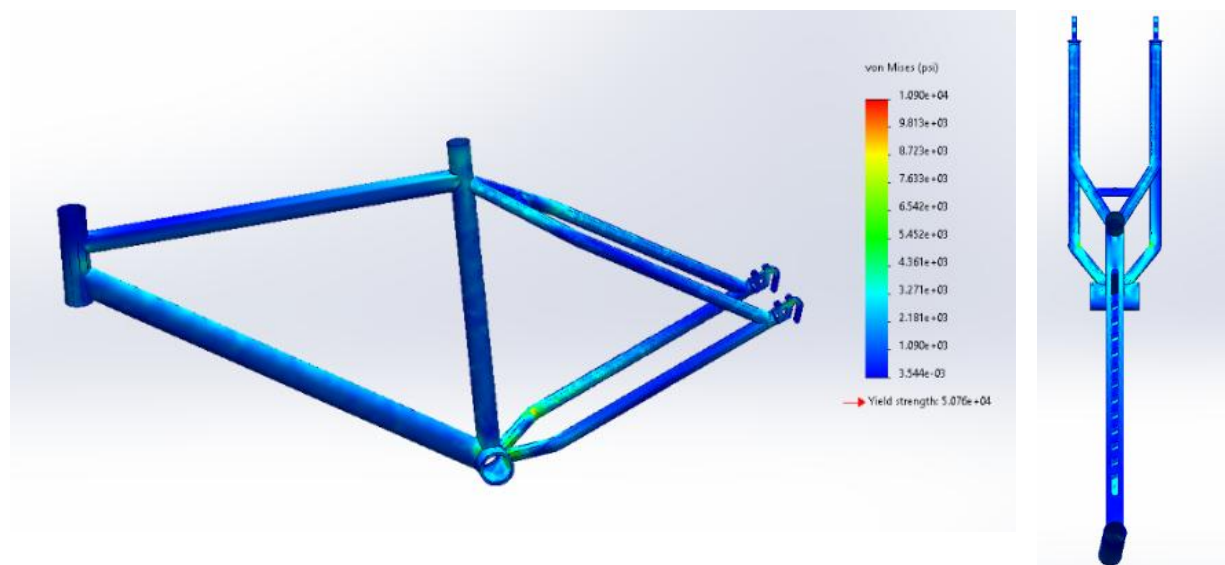


Table 3.

Comparison of Mechanical Results Between Original and Redesigned Bicycle Frame

Property	Original Design	New Design	Total Change
Frame Weight (lbf)	3.03	2.68	-0.35 (-11.55%)

Cross-Sectional Area (in ²)			
Top Tube	0.11	0.0875	-0.0225
Down Tube	0.12	0.1	-0.02
Yield Factor of Safety			
Top Tube	21.82	17.36	-4.46
Down Tube	20.53	17.12	-3.41
Fatigue Factor of Safety			
Down Tube	24.61	20.51	-4.1
Failure Weights (lbf)			
Yield	1300	1295	-5
Fatigue	1250	1250	0

Rationale

The redesigned frame achieves a greater than 10% weight reduction while maintaining structural integrity under both static and fatigue loading conditions. This was accomplished by introducing cutouts in the down tube and reducing the wall thickness of the top and down tubes—two members identified as having lower load demands. Finite Element Analysis (FEA) and spreadsheet-based safety factor calculations confirm that these changes do not result in significant local failure or compromise overall durability significantly. The updated design still maintains the conservative safety margins typical in bicycle

engineering while introducing a visually distinct, stylized aesthetic. This innovation balances weight reduction and safety while offering the possibility for customization potential that appeals to cycling enthusiasts, particularly Gen-Z, who enjoy personalizing their gear.

Reflection

This project reinforced key concepts from the course, including identifying the critical stress element, analyzing combined loading scenarios (axial, torsion, and bending), and evaluating both static and fatigue failure using appropriate failure theories. It also emphasized the importance of applying design criteria to ensure safety and performance under real-world conditions. These tasks strengthened essential engineering skills such as problem-solving, critical thinking, and interpreting analysis results in the context of a physical design.

One of the most challenging aspects of the project was translating theoretical knowledge into a tangible and realistic model (a bicycle)—moving beyond the idealized beams and simplified scenarios encountered in engineering coursework. What I appreciated most was the opportunity to apply the engineering theory I was learning to a real, practical problem. The hands-on experience with 3D printing was especially enjoyable, as it introduced me to a new manufacturing process I hadn't used before. It also allowed me to refine my SolidWorks skills and personalize my design—both technically and aesthetically—through finishing techniques like spray painting.

While the project was rewarding, it required a substantial amount of work outside of class for a major portion of the final grade. One suggestion for improvement would be to incorporate more project-based

class time or guided homework to support timely progress with the project and better alignment with the topics being studied in real time. This could help bridge the gap between theoretical analysis and practical design implementation and help to reinforce key course concepts.

References

Nisbett, J. Keith. *Shigley's Mechanical Engineering Design*. 12th ed., McGraw-Hill Education, 2024.

Appendix A

A-1

Maximum Weight Calculations

find maximum weight for infinite life (10^6 cycles)

find endurance limit

$$S_e = K_a \times K_b \times K_c \times K_d \times K_e \times K_f \times S_e'$$

$$K_d = K_e = 1, \quad K_c = 0.85 \text{ (axial)}$$

$$K_a = a S_{ut}^b$$

$$= 2.70(60.9)^{-0.265} \quad \text{from Table 6-12 (cold drawn)}$$

$$= 0.9087 \approx 0.91$$

$$K_b = 0.879(d)^{-0.107} \quad \text{for } (0.11 \leq d \leq 2 \text{ in})$$

$$= 0.879(0.75)^{-0.107}$$

$$= 0.9064 \approx 0.91$$

$$K_c = 0.85 \text{ (axial only)}$$

$$K_f = 1 + q(K_t - 1) = 1 \quad \text{No notches, uniform cross-sectional area}$$

$$S_e = K_a \times K_b \times K_c \times K_d \times K_e \times K_f \times S_e'$$

$$= 0.91 \times 0.91 \times 0.85 \times 1 \times 1 \times 0.5 \times 60.9 = 21.43 \text{ ksi}$$

$$\sigma_{min} = 0 \quad \sigma_{max} = \frac{F}{A} \quad \rightarrow \quad \sigma_a = \left| \frac{\sigma_{max} - \sigma_{min}}{2} \right|$$

$$\sigma_m = \frac{\sigma_{max} + \sigma_{min}}{2}$$

$$\text{Goodman: } \frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} \leq 1 \text{ for fatigue failure}$$

* use spreadsheet to calculate
max weight

A-2

Max Drop Distance Calculations

Find max drop distance for infinite life

Known Values (Material properties and Stage 1/2 calculations):

$$S_{ut} = 60900$$

$$R = 47.76 \text{ lbf}$$

$$W = 200 \text{ lbf}$$

$$F_{\text{impact}} = 997\sqrt{h}$$

$$\sigma_{\text{bend rider}} = 935.092 \text{ psi}$$

$$\tau_{\text{torsion rider}} = 979.659 \text{ psi}$$

} from stage 2
@ 200 lbf rider weight

$$K_a = K_b = 0.91 \leftarrow \begin{array}{l} \text{from frame analysis} \\ \text{(fork is same material)} \end{array}$$

find endurance limit

$$\begin{aligned} S_e &= K_a \times K_b \times K_c \times K_d \times K_e \times S_e' \\ &= 0.91 \times 0.91 \times 1 \times 1 \times 1 \times 0.5 \times 60800 \\ &= 25215.645 \text{ psi} \end{aligned}$$

find σ'_a and σ'_m

K_F/K_T :



fork / steeler
interface

$$D/d = \frac{1.5 \text{ in}}{1.00 \text{ in}} = 1.5$$

$$r/d = \frac{0.1 \text{ in}}{1 \text{ in}} = 0.1$$

→ No axial (per stage 2)

figure A-15-8

$$K_{t, \text{torsion}} = 1.45$$

figure A-15-9

$$K_{t, \text{bend}} = 1.7$$

Welded → $K_F = K_T$

$$\sigma_{\text{bend max}} = \frac{\text{Force}}{R} \times \sigma_{\text{bend rider}} = \frac{997\sqrt{h} \text{ lbf}}{47.76 \text{ lbf}} \times 935.092 \text{ psi} = 19520\sqrt{h}$$

$$\tau_{\text{Torsion max}} = \frac{F_{\text{impact}}}{R} \times \tau_{\text{torsion rider}} = \frac{997\sqrt{h} \text{ lbf}}{47.76 \text{ lbf}} \times 979.659 \text{ psi} = 20451\sqrt{h}$$

$$\sigma'_a \text{ bend} = \frac{\sigma_{\text{bend max}} - \sigma_{\text{bend min}}}{2} = \frac{19520\sqrt{h} - 935.092 \text{ psi}}{2}$$

$$\sigma'_m \text{ bend} = \frac{\sigma_{\text{bend max}} + \sigma_{\text{bend min}}}{2} = \frac{19520\sqrt{h} + 935.092 \text{ psi}}{2}$$

Continued on next page

$$\tau_{a \text{ torsion}} = \frac{\tau_{\max} - \tau_{\min}}{2} = \frac{20451 \sqrt{h} - 919.659 \text{ psi}}{2}$$

$$\tau_{m \text{ torsion}} = \frac{\tau_{\max} + \tau_{\min}}{2} = \frac{20451 \sqrt{h} + 919.659 \text{ psi}}{2}$$

$$\begin{aligned} \sigma'_a &= \left[\left[(Kf_{\text{bend}}) \sigma_{\text{bend}} + (Kf_{\text{axial}}) \frac{\sigma_{\text{axial}}}{0.85} \right]^2 + 3 \left[(Kf_{\text{torsion}}) \tau_{a \text{ torsion}} \right]^2 \right]^{1/2} \\ &= \left[\left[(1.7) \left(\frac{19520 \sqrt{h} - 935.092}{2} \right) + 0 \right]^2 + 3 \left[(1.45) \left(\frac{20451 \sqrt{h} - 919.659}{2} \right) \right]^2 \right]^{1/2} \\ &= \left[(16592 \sqrt{h} - 794.828)^2 + 3 (14027 \sqrt{h} - 724.726)^2 \right]^{1/2} \end{aligned}$$

$$\begin{aligned} \sigma'_m &= \left[\left[(Kf_{\text{bend}}) \sigma_{\text{bend}} + (Kf_{\text{axial}}) \sigma_{\text{axial}} \right]^2 + 3 \left[(Kf_{\text{torsion}}) \tau_{m \text{ torsion}} \right]^2 \right]^{1/2} \\ &= \left[\left[(1.7) \left(\frac{19520 \sqrt{h} + 935.092}{2} \right) + 0 \right]^2 + 3 \left[(1.45) \left(\frac{20451 \sqrt{h} + 919.659}{2} \right) \right]^2 \right]^{1/2} \\ &= \left[(16592 \sqrt{h} + 794.828)^2 + 3 (14027 \sqrt{h} + 724.726)^2 \right]^{1/2} \end{aligned}$$

find h (using Goodman)

$$\text{Goodman: } \frac{\sigma'_a}{s_e} + \frac{\sigma'_m}{s_{ut}} = 1$$

$$\frac{[(16592 \sqrt{h} - 794.828)^2 + 3(14027 \sqrt{h} - 724.726)^2]^{1/2}}{25215.645 \text{ psi}} + \frac{[(16592 \sqrt{h} + 794.828)^2 + 3(14027 \sqrt{h} + 724.726)^2]^{1/2}}{60900 \text{ psi}} = 1$$

→ using MATLAB → $h = 0.405 \text{ in}$

Appendix B

B-1

Excel Macro – Increment Rider Weight until Fatigue Failure

Module1 - 1

```
Sub FindCriticalFatigueWeight()  
  
    Dim RiderWeight As Double  
    Dim WeightIncrement As Double  
    Dim MaxIterations As Long  
    Dim Iteration As Long  
    Dim FOSRange As Range  
    Dim FOSCell As Range  
  
    ' Parameters  
    RiderWeight = 200  
    WeightIncrement = 1  
    MaxIterations = 10000  
    Set FOSRange = Sheets("Static Analysis Calculator").Range("G32:G36")  
  
    'Increment weight until Fatigue FOS <1 reached  
    For Iteration = 1 To MaxIterations  
  
        ' Update weight in cell C3  
        Sheets("Static Analysis Calculator").Range("C3").Value = RiderWeight  
  
        ' Check Fatigue FOS values  
        For Each FOSCell In FOSRange  
            If IsNumeric(FOSCell.Value) And FOSCell.Value < 1 Then  
                Exit Sub      ' Stop if any FOS is < 1  
            End If  
        Next FOSCell  
  
        ' Increment weight  
        RiderWeight = RiderWeight + WeightIncrement  
  
    Next Iteration  
  
End Sub
```

B-2

MATLAB Code – Solving for Max Drop Height

```
% Clear workspace, command window, and close all figures
clear; clc; close all;

% Define symbolic variable
syms h positive

% Material Properties
Se = 27410;      % Endurance limit (psi)
Sut = 60900;     % Ultimate tensile strength (psi)

% Fatigue Stress Concentration Factors
kf_bending = 1.7;      % For bending
kfs_torsion = 1.45;    % For torsion

% Bending stresses
sig_a_bend = (19520*sqrt(h) - 935.092)/2;
sig_m_bend = (19520*sqrt(h) + 935.092)/2;

% Torsional stresses
tau_a = (20451*sqrt(h) - 999.659)/2;
tau_m = (20451*sqrt(h) + 999.659)/2;

% Effective Stresses (alternating and mean)
sig_a_prime = sqrt((kf_bending * sig_a_bend)^2 + 3 * (kfs_torsion * tau_a)^2);
sig_m_prime = sqrt((kf_bending * sig_m_bend)^2 + 3 * (kfs_torsion * tau_m)^2);

% Goodman Equation
Goodman_fatigue = (sig_a_prime / Se) + (sig_m_prime / Sut) == 1;

% Solve for h
h_sol = solve(Goodman_fatigue, h);

% Convert to numeric
h_numeric = double(h_sol);

% Result
fprintf('Maximum drop height for infinite fatigue life: %.3f inches\n',
h_numeric);

Maximum drop height for infinite fatigue life: 0.405 inches
```

Published with MATLAB® R2024b

B-3

Mesh Convergence Graph

